Detector-Guided Vocoder Training for Adversarial Text-to-Speech Synthesis

Jiayan Cui, Davide Salvi, Paolo Bestagini, Stefano Tubaro

Dipartimento di Elettronica, Informazione e Bioingegneria (DEIB), Politecnico di Milano
jiayan.cui@mail.polimi.it, {davide.salvi, paolo.bestagini, stefano.tubaro}@polimi.it

***Abstract*—In this paper, we propose a novel Text-to-Speech (TTS) framework designed to perform adversarial attacks against speech deepfake detection systems. Given a TTS model and a detector capable of classifying its outputs as synthetic, our approach fine-tunes the vocoder by integrating the detector into the TTS training loop. Unlike traditional adversarial methods that rely on post-hoc perturbations or external noise injection, our approach enables the generation of synthetic speech that is intrinsically adversarial, capable of lowering the target detector's reliability without requiring additional post-processing steps. We evaluate the framework in both white-box and black-box settings, showing its effectiveness against known and unseen detectors. We also propose a multi-detector strategy that allows the system to simultaneously learn to deceive multiple detectors, showcasing the scalability of the method. Experimental results confirm the efficacy of our approach, which has the potential to impact forensic security and detection countermeasures. At the same time, subjective listening tests reveal that the proposed method also enhances synthesized speech quality.**

***Index Terms*—Audio Forensics, Speech Deepfake, Adversarial Attacks, Anti-forensics, Speech Synthesis**

I. Introduction

Recent advances in speech synthesis and deep learning have led to significant progress in text-to-speech (TTS) technologies, enabling the generation of highly realistic speech with minimal human effort [1], [2], [3]. These technologies have become increasingly important for applications such as accessibility, entertainment, education, and human-computer interaction. At the same time, however, they have also facilitated the creation of speech deepfakes, i.e., synthetic speech signals that closely resemble authentic human speech and can impersonate a target speaker or convey arbitrary spoken content. The increasing realism of these systems has raised significant concerns regarding misinformation, fraud, identity theft, and other malicious applications, motivating extensive research on reliable speech deepfake detection.

To address these threats, the multimedia forensics community has developed a wide range of speech deepfake detectors. Most of these rely on supervised learning, ranging from the use of traditional signal processing techniques [4], [5], to that of more sophisticated models, such as Graph Attention Networks [6] and Transformers [7], [8]. Other strategies exploit transfer learning from pre-trained systems [9], [10], or analyze semantic inconsistencies in the signals [11], [12] to determine their authenticity.

In parallel with the development of forensic detection tools, there has been growing interest in anti-forensic techniques. These aim to identify and exploit vulnerabilities in detection systems by designing adversarial attacks [13], [14]. In the audio domain, adversarial attacks have predominantly been implemented as post-processing operations. These involve injecting learned additive noise or perturbations into audio signals to hinder their forensic traces and deceive the detectors that are used to process them [15], [16], [17], [18]. For instance, [19] proposed an attack that injects convolutional noise using a linear time-invariant filter, while [20] developed a noise injection attack based on a generalized Hammerstein model to target specific speech deepfake detectors. Similarly, [21] presented a joint adversarial attack capable of deceiving both Automatic Speaker Verification (ASV) systems and speech deepfake detectors simultaneously, while [22] proposed a novel attack in a physical access context, where a replay process is involved. Despite their efficacy, the dependence on post-processing is a key limitation of these methods. Adversarial processing remains a separate stage of the generation pipeline, introducing additional computational complexity and potentially leaving artifacts that can affect perceptual quality or be exploited by more robust forensic methods.

In this paper, we depart from the paradigm of additive noise-based attacks and propose a novel approach that integrates adversarial perturbation directly into the speech synthesis process. Specifically, we fine-tune the vocoder of a TTS system by incorporating the loss of one or more speech deepfake detectors into the vocoder training objective. During optimization, the detector provides differentiable feedback that guides the vocoder towards generating speech signals that retain high perceptual quality while being classified as authentic by the target detector. As a result, the generated speech is intrinsically adversarial and does not require any additional waveform perturbation or post-processing at inference time.

The primary contributions of this work are the following:

- *Detector-guided adversarial synthesis* – We propose a detector-guided vocoder optimization framework that generates intrinsically adversarial speech without requiring post-processing operations.
- *Real-time applicability* – Since adversarial fine-tuning occurs offline, the system generates adversarial speech in real time without additional runtime overhead.
- *Generalization capability* - We evaluate the method under white-box and black-box conditions, showing that it effectively deceives both targeted and unseen detectors. Furthermore, we introduce a multi-detector strategy, enabling

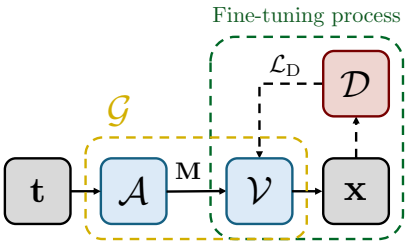


Fig. 1: Pipeline of the proposed TTS adversarial model.

simultaneous attacks on multiple architectures.

- *Enhanced speech quality* – Beyond adversarial robustness, our fine-tuning improves perceptual quality, as validated through a subjective listening test.

II. Proposed Method

In this paper, we propose an adversarial TTS framework that generates speech intrinsically optimized to evade speech deepfake detectors. Rather than applying adversarial perturbations after synthesis, our approach incorporates the detector directly into the vocoder optimization process, enabling the generator to learn synthesis strategies that reduce the detector confidence while preserving speech quality.

A. Problem Formulation

The speech deepfake detection problem can be formally defined as follows. Let $\mathbf{x} \in \mathbb{R}^T$ denote a discrete-time speech signal with class label $y \in \{0, 1\}$, where $y = 0$ indicates an authentic signal and $y = 1$ a synthetically generated one. The goal of this task is to develop a speech deepfake detector $\mathcal{D} : \mathbb{R}^T \to [0, 1]$ able to estimate the class of $\mathbf{x}$ as $\hat{y} = \mathcal{D}(\mathbf{x})$, where $\hat{y}$ represents the likelihood of $\mathbf{x}$ being synthetic.

The goal of an adversarial attack is to generate a signal $\tilde{\mathbf{x}}$ that is perceptually similar to $\mathbf{x}$ but induces an incorrect prediction by the target detector $\mathcal{D}$, such as

$$\mathcal{D}(\tilde{\mathbf{x}}) < \mathcal{D}(\mathbf{x}) \quad \text{subject to} \quad \mathbf{x} \approx \tilde{\mathbf{x}}, \tag{1}$$

where $\approx$ indicates perceptual similarity.

B. Adversarial TTS Architecture

Let us define a TTS generator $\mathcal{G}$ as a system that receives a text string $\mathbf{t}$ as input and returns a speech signal $\mathbf{x}$ as output, as in $\mathbf{x} = \mathcal{G}(\mathbf{t})$. We consider the TTS generator $\mathcal{G}$ as the composition of two cascade modules:

- An *Acoustic Model* $\mathcal{A}$, that maps text $\mathbf{t}$ to mel-spectrogram $\mathbf{M} \in \mathbb{R}^{F \times N}$ through

$$\mathbf{M} = \mathcal{A}(\mathbf{t}; \theta_{\mathcal{A}}), \tag{2}$$

  where $\theta_{\mathcal{A}}$ denotes model parameters, F the mel-frequency bins, and N the temporal frames.
- A *Vocoder* $\mathcal{V}$, which synthesizes $\mathbf{M}$ into a T-samples waveform $\mathbf{x} \in \mathbb{R}^T$ via

$$\mathbf{x} = \mathcal{V}(\mathbf{M}; \theta_{\mathcal{V}}). \tag{3}$$

The resultant speech sample is defined as $\mathbf{x} = \mathcal{V}(\mathcal{A}(\mathbf{t}))$.

Since $\mathbf{x}$ is synthetically generated, its class label is $y = 1$, and an ideal detector should assign it a high $\hat{y}$ value, meaning a high probability of belonging to the synthetic class. Our objective is therefore to adapt $\mathcal{G}$ so that the synthesized speech $\tilde{\mathbf{x}}$ is instead classified as authentic. Rather than retraining the entire TTS system, we keep the acoustic model $\mathcal{A}$ fixed and fine-tune only the vocoder $\mathcal{V}$. This choice preserves the linguistic and speaker representations learned by the acoustic model while allowing the waveform generation process to suppress detector-relevant forensic cues.

To do so, we optimize the vocoder parameters $\theta_{\mathcal{V}}$ using a composite loss that combines the original vocoder training objective $\mathcal{L}_{\mathrm{V}}$ with an adversarial detection loss $\mathcal{L}_{\mathrm{D}}$. The original vocoder loss preserves the synthesis quality learned during standard training, whereas the adversarial term encourages the generated waveform to be classified as authentic by the target detector. Importantly, during this optimization, the parameters of the detector $\mathcal{D}$ remain frozen and gradients are back-propagated only through the detector to update the vocoder. The overall loss is therefore defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\mathrm{V}} + \lambda \mathcal{L}_{\mathrm{D}}, \tag{4}$$

where λ is a weighting factor. The detection loss $\mathcal{L}_{\mathrm{D}}$ is computed by evaluating the synthesized waveform with the target detector and encouraging the detector to predict the authentic class. Formally, it is defined as:

$$\mathcal{L}_{\mathrm{D}} = \mathbb{E}_{\mathbf{t}} \left[-\log \left(1 - \mathcal{D} \left(\mathcal{V}(\mathcal{A}(\mathbf{t})) \right) \right) \right]. \tag{5}$$

This corresponds to minimizing the binary cross-entropy with respect to the authentic class. Consequently, the gradients propagated through the detector encourage the vocoder to generate waveforms whose forensic characteristics resemble those of genuine speech, while the original vocoder objective prevents the optimization from degrading synthesis quality.

The proposed formulation naturally extends to multiple target detectors. Instead of optimizing against a single detector, $\mathcal{L}_{\mathcal{D}}$ can be expressed as a weighted combination of detector-specific losses:

$$\mathcal{L}_{\mathrm{D}} = \sum_{k=1}^{K} \alpha_k \mathbb{E}_{\mathbf{t}} \left[-\log \left(1 - \mathcal{D}_k \left(\mathcal{V}(\mathcal{A}(\mathbf{t})) \right) \right) \right], \tag{6}$$

where $\mathcal{D}_k$ denotes the k-th detector and α_k weights detector-specific objectives. By minimizing the composite objective, the vocoder jointly optimizes speech fidelity and adversarial effectiveness, producing synthetic speech that preserves perceptual quality while reducing the confidence of one or more target speech deepfake detectors. Figure 1 shows the complete adversarial TTS pipeline.

III. Experimental Setup

In this section, we describe the experimental setup adopted to evaluate the proposed framework. We first present the considered synthesis and detection systems, followed by the training procedure and the datasets used throughout the experiments.

TABLE I: Vocoder configurations considered throughout the experiments.

Name	Detector-guided training	Purpose
$\mathcal{V}$	None	Baseline synthesis
$\mathcal{V}^{\mathrm{R}}$	RawNet2	Single-detector attack
$\mathcal{V}^{\mathrm{RT}}$	RawNet2 + TSSD	Multi-detector attack

A. Considered Systems

The proposed framework is built upon a two-stage TTS pipeline composed of an acoustic model and a neural vocoder. As acoustic model $\mathcal{A}$, we employ an enhanced version of FastSpeech2 [3], incorporating the modifications proposed in [23], [24], [25] to improve speaker conditioning and synthesis quality. Specifically, the model integrates Utterance- and Phoneme-Level Encoders, pyannote-based speaker embeddings [24], a Conditional Layer for speaker adaptation, and Random Window Discriminators (RWDs) [25]. We refer the reader to the corresponding references for a detailed description of these components.

As vocoder $\mathcal{V}$, we adopt BigVGAN [26], a state-of-the-art neural vocoder for waveform synthesis. The acoustic model remains fixed throughout our experiments, while only the vocoder is fine-tuned using the detector-guided objective described in Section II. This design isolates the effect of adversarial optimization on waveform generation while preserving the linguistic and speaker representations learned by the acoustic model.

To assess the impact of detector-guided optimization, we train three different vocoder variants:

- $\mathcal{V}$: The original vocoder, trained only with the original loss function $\mathcal{L}_{\mathrm{V}}$ proposed in [26], without any adversarial loss.
- $\mathcal{V}^{\mathrm{R}}$: Trained including $\mathcal{L}_{\mathrm{D}}$ with $K = 1$, to fool a single detector (RawNet2).
- $\mathcal{V}^{\mathrm{RT}}$: Trained with a multi-detector loss $\mathcal{L}_{\mathrm{D}}$, aiming to deceive the $K = 2$ detectors together (RawNet2 and TSSD).

Since detector-guided optimization is performed exclusively during offline training, the computational cost is incurred only once during fine-tuning. At inference time, the proposed framework has the same computational complexity and latency as the original BigVGAN vocoder, as no additional detector inference or waveform post-processing is required.

Finally, we evaluate our adversarial framework against three end-to-end speech deepfake detectors $\mathcal{D}$ that operate directly on raw audio waveforms: RawNet2 [27], TSSD [28], and AASIST [6]. RawNet2 and TSSD are used both during adversarial optimization and evaluation, representing the white-box attack scenario. In contrast, AASIST is never involved during training and is used exclusively during testing to assess the transferability of the proposed attack under black-box conditions. The three vocoder configurations considered throughout the paper are summarized in Table I.

B. Training parameters

The acoustic model was trained following the original FastSpeech2 implementation, adopting the optimization strategy described in [3]. Likewise, the baseline BigVGAN vocoder was trained using the configuration proposed in [26]. Unless otherwise specified, we retain the original training hyperparameters of both architectures.

The adversarial fine-tuning stage differs only in the optimization objective. Starting from the baseline vocoder, we optimize the parameters according to the composite loss introduced in Eq. 4, combining the original vocoder objective with the detector-guided loss. During this phase, the acoustic model and the detector networks remain frozen, and gradients are back-propagated only to update the vocoder parameters.

In practice, Eq. 5 is implemented as the binary cross-entropy loss between the detector prediction and the authentic class, as in

$$\mathcal{L}_{\mathrm{D}} = \lambda \cdot \mathrm{CE}(\mathcal{D}(\mathbf{x}), 0), \tag{7}$$

where the target label 0 denotes authentic speech and $\lambda = 0.05$ is a scaling constant that controls the contribution of the adversarial objective. Minimizing this loss encourages the vocoder to generate speech that is increasingly difficult for the target detector to classify as synthetic.

All detectors were trained under identical conditions using the Adam optimizer with cross-entropy loss. Additional implementation details, including optimizer parameters and learning-rate schedules, follow the original implementations of the corresponding models. All experiments were conducted on a workstation equipped with an Intel i9-14900K CPU, 64 GB RAM, and an NVIDIA RTX 4090 GPU. The code used for training and evaluating the models, along with audio samples generated by the proposed TTS systems, is accessible at this link[1].

C. Datasets

We trained the acoustic model $\mathcal{A}$ and the vocoder $\mathcal{V}$ using LibriTTS [29], a dataset derived from LibriSpeech [30] and specifically processed for TTS applications. This provides approximately 260 hours of high-quality speech from 1158 speakers sampled at $f_{\mathrm{s}} = 22.05\,\mathrm{kHz}$. Following the preprocessing adopted in our experiments, utterances shorter than $3\,\mathrm{s}$ and recordings flagged for quality issues were discarded. The resulting dataset was randomly split into 90% for training and 10% for validation.

To fine-tune BigVGAN, we considered both real and synthetic inputs, utilizing mel-spectrograms extracted from LibriTTS signals, as well as those generated by FastSpeech2. This approach adapts the vocoder to the specific characteristics of the output of the acoustic model, resulting in a more stable and coherent speech generation.

We trained the deepfake detectors $\mathcal{D}$ using an internal dataset balanced between real and synthetic speech for a total of $251\,154$ tracks sampled at $f_{\mathrm{s}} = 22.05\,\mathrm{kHz}$. Authentic recordings were drawn from LibriTTS, whereas synthetic speech was generated using the baseline TTS pipeline employing the vocoder $\mathcal{V}$.

[1] https://github.com/polimi-ispl/adversarial-tts/

TABLE II: Detection performance of the considered speech deepfake detectors $\mathcal{D}$ against different iterations of the vocoder $\mathcal{V}$.

	RawNet2			TSSD			AASIST		
	EER	Bal. Acc.	AUC	EER	Bal. Acc.	AUC	EER	Bal. Acc.	AUC
$\mathcal{V}$	5.8%	94.15%	97.08%	3.40%	95.73%	98.80%	0.85%	99.16%	99.95%
$\mathcal{V}^{\mathrm{R}}$	39.05%	60.85%	64.58%	38.33%	60.54%	66.08%	21.09%	78.91%	87.07%
$\mathcal{V}^{\mathrm{RT}}$	43.87%	55.17%	57.72%	43.31%	57.03%	59.74%	23.26%	76.75%	84.55%

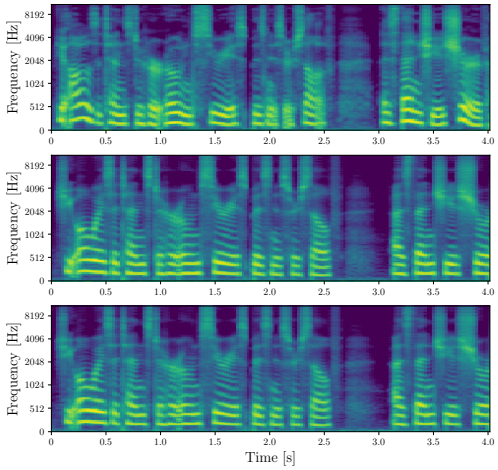


Fig. 2: Mel-spectrograms of real and synthetic signals. The real signal (top) is from LibriTTS, the synthetic ones are from $\mathcal{V}$ (middle) and $\mathcal{V}^{\mathrm{R}}$ (bottom).

To evaluate the effectiveness of detector-guided vocoder training, we constructed three test datasets that share the same authentic speech recordings while differing only in the synthetic samples. Specifically, the fake speech in each dataset was generated using one of the three vocoder variants ($\mathcal{V}$, $\mathcal{V}^{\mathrm{R}}$, and $\mathcal{V}^{\mathrm{RT}}$). This controlled protocol isolates the effect of adversarial fine-tuning by ensuring that the authentic data remain identical across all evaluation scenarios.

Figure 2 presents the mel-spectrograms of a real test track alongside its synthetic counterparts from $\mathcal{V}$, and $\mathcal{V}^{\mathrm{R}}$, showing that the differences between them are almost imperceptible.

IV. Results

In this section, we evaluate the proposed framework from three complementary perspectives. First, we assess its ability to degrade speech deepfake detectors under both white-box and black-box conditions. Second, we investigate whether jointly optimizing against multiple detectors improves attack transferability. Finally, we evaluate the perceptual quality of the generated speech using both objective and subjective metrics.

Detection performance is measured in terms of Area Under the Curve (AUC), Balanced Accuracy, and Equal Error Rate (EER). While ideal detector performance corresponds to high AUC and Balanced Accuracy and low EER, the objective of the proposed adversarial framework is the opposite: to reduce the detector's ability to discriminate between authentic and synthetic speech. Since only the synthetic signals are modified while the authentic ones remain unchanged, the theoretical optimum of a successful attack corresponds to 50% for all three metrics, indicating random discrimination between the two classes.

We evaluate the three speech deepfake detectors (RawNet2, TSSD, and AASIST) using speech synthesized by the three vocoder configurations introduced in Section III, namely the baseline vocoder $\mathcal{V}$, the single-detector adversarial vocoder $\mathcal{V}^{\mathrm{R}}$, and the multi-detector adversarial vocoder $\mathcal{V}^{\mathrm{RT}}$. This experimental protocol enables us to separately evaluate baseline detector performance, white-box attack effectiveness, and transferability to previously unseen detector architectures.

Table II summarizes the detection performance obtained under the three evaluation settings. As expected, all detectors achieve excellent performance on speech generated by the baseline vocoder $\mathcal{V}$, with AUC values above 97% and EER below 6%. Since these synthetic samples closely match those used during detector training, this experiment establishes a strong reference point for assessing the effectiveness of the proposed adversarial optimization. Among the evaluated systems, AASIST achieves the highest detection accuracy, although all three detectors reliably distinguish authentic from synthetic speech.

Detector-guided fine-tuning substantially degrades the performance of the targeted detectors. When optimizing exclusively against RawNet2, its EER increases from 5.8% to 39.1%, approaching the theoretical upper bound of 50% associated with random guessing. Interestingly, TSSD also experiences a comparable degradation (EER$=38.3\%$) despite not being explicitly optimized during training, suggesting that the learned adversarial modifications suppress forensic cues that are not entirely detector-specific. Even the unseen AASIST detector exhibits a marked reduction in performance, with its EER increasing to 21.1%, indicating that the proposed optimization already provides a significant degree of transferability beyond the targeted architecture.

Joint optimization against RawNet2 and TSSD further improves the attack. Both targeted detectors exhibit EER values above 43%, while their Balanced Accuracy decreases to approximately 56%, indicating detector performance close to random discrimination. More importantly, the performance of AASIST also decreases further, reaching an EER of 23.3% despite never being involved during adversarial optimization. This result indicates that jointly optimizing against multiple detectors improves attack transferability by encouraging the vocoder to suppress forensic cues shared across different

detector architectures rather than overfitting to a single target model.

Figure 3 shows the detection performance of RawNet2 when tested against data generated by $\mathcal{V}^{\text{RT}}$, via a confusion matrix. Almost all the synthetic tracks are misclassified as real, with an exceptionally high True Negative Rate (TNR) (0.946) and low True Positive Rate (TPR) (0.073), aligning with our adversarial goal. With a balanced accuracy of 55%, the proposed attack effectively reduces the detector's performance to near-random guessing.

Beyond evaluating adversarial effectiveness, we also investigate the perceptual quality of the generated speech. We hypothesize that detector-guided optimization not only suppresses detector-relevant forensic traces but also reduces synthesis artifacts that distinguish generated speech from authentic recordings. To test this hypothesis, we first compute the Fréchet Audio Distance (FAD) [31] between authentic speech and the three synthetic datasets using CLAP [32] embeddings. The FAD decreases consistently from 31.25 for the baseline vocoder $\mathcal{V}$ to 22.95 for $\mathcal{V}^{\text{R}}$ and 17.81 for $\mathcal{V}^{\text{RT}}$, corresponding to an improvement of approximately 25% at each optimization stage. This progressive reduction suggests that adversarial fine-tuning moves the distribution of synthesized speech closer to that of authentic recordings while simultaneously improving resistance to forensic detection.

To complement the objective evaluation, we conducted a MUSHRA-like perceptual listening test with 20 expert listeners to assess the level of realism of the generated tracks. The study was anonymous and GDPR-compliant, with informed consent obtained and no personal data collected. Figure 4 presents the results of this analysis, indicating that the fine-tuned vocoders $\mathcal{V}^{\text{R}}$ and $\mathcal{V}^{\text{RT}}$ were perceived as more realistic compared to the initial $\mathcal{V}$ vocoder, progressively approaching the quality of real audio. Together with the FAD analysis, these results indicate that our adversarial fine-tuning not only enhances attack robustness but also increases the perceptual quality of synthesized speech, making it a promising strategy for advancing adversarial TTS systems.

The observed improvement in perceptual quality is particularly interesting because the detector loss is never optimized explicitly for human perception. A possible explanation is that speech deepfake detectors rely on subtle synthesis artifacts that are also perceptually undesirable. By encouraging the vocoder to suppress these detector-relevant cues while preserving the original synthesis objective, the optimization may simultaneously improve detector evasion and speech naturalness. Although further investigation is required to validate this hypothesis, the consistent improvements observed in both FAD and the listening test suggest a strong relationship between forensic artifacts and perceptual quality.

Overall, the experiments demonstrate three main findings. First, detector-guided vocoder optimization substantially degrades the performance of targeted speech deepfake detectors under white-box conditions. Second, jointly optimizing against multiple detectors improves transferability to previously unseen detector architectures. Finally, adversarial fine-tuning

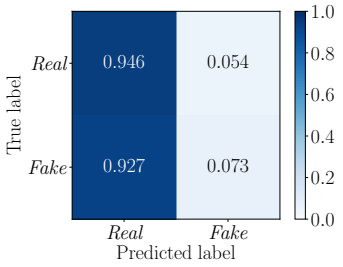


Fig. 3: Confusion matrix obtained by testing RawNet2 against the test dataset containing fake tracks generated by the vocoder $\mathcal{V}^{\text{RT}}$.

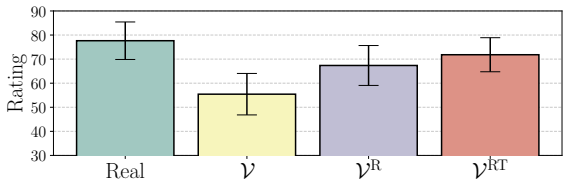


Fig. 4: Results of the perceptual listening test evaluating the realism of the real and synthetic tracks generated with different iterations of the vocoder $\mathcal{V}$.

unexpectedly improves both objective and subjective measures of speech quality, indicating that suppressing detector-relevant forensic cues can also increase the perceptual realism of synthesized speech.

V. Conclusions

In this paper, we presented a detector-guided TTS framework capable of generating speech that is intrinsically adversarial to speech deepfake detectors. By integrating one or more frozen detector models into the vocoder optimization process, the proposed approach directly learns waveform generation strategies that reduce detector confidence while preserving the perceptual quality of the synthesized speech. Unlike existing adversarial attacks that rely on post-processing perturbations, the proposed framework produces adversarial speech during synthesis, eliminating the need for any additional processing at inference time.

Extensive experiments against three state-of-the-art speech deepfake detectors showed the effectiveness of the proposed approach under both white-box and black-box conditions. Detector-guided optimization substantially degraded the performance of the targeted detectors, while joint optimization against multiple detectors further improved transferability to previously unseen architectures. Interestingly, objective and subjective evaluations consistently showed that adversarial fine-tuning also enhances the perceptual realism of the generated speech, suggesting that suppressing detector-relevant forensic cues may simultaneously reduce synthesis artifacts.

The proposed framework also highlights the growing challenge of securing speech synthesis systems against increasingly sophisticated adversarial attacks. Although our objective

is to expose weaknesses in current forensic methods and support the development of more robust countermeasures, the same techniques could potentially be misused to generate synthetic speech that is more difficult to detect. We therefore advocate the use of this work exclusively for defensive security research and encourage continued investigation into robust detection methods, benchmarking protocols, and ethical guidelines for generative audio technologies.

While the proposed framework demonstrates strong performance against three representative speech deepfake detectors, the evaluation is limited to a single TTS architecture and a controlled synthesis scenario. Future work will investigate additional vocoders, multilingual datasets, real-world transmission conditions, and more computationally efficient optimization strategies, with the ultimate goal of developing speech deepfake detectors that remain robust against adversarially trained synthesis models.

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